<u>Fake or Reyal: AI-Based Document Verification System</u>

<u>(Fake Detection)</u>

A Thesis Project presented to the

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Bachelor of Science in Computer Science

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ABSTRACT

Title : Fake or Reyal: AI-Based Document Verification System (Fake Detection)

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Project Description : The increasing use of digital documents in academic, business, and government transactions has led to a growing risk of document falsification and fraud. Traditional methods of document verification are often manual, time-consuming, and prone to human error. This study proposes the development of a system entitled "Fake or Real: AI-Based Document Verification System", which utilizes artificial intelligence to automatically detect fraudulent or tampered documents.

The system employs machine learning and image processing techniques to analyze key features of documents such as text patterns, signatures, layouts, and embedded security elements. By training the model using datasets of authentic and fake documents, the system can classify whether a submitted document is genuine or suspicious. The system

aims to provide fast, accurate, and reliable verification results, reducing the need for manual validation.

This study is significant for institutions such as schools, companies, and government agencies that frequently handle sensitive documents. The proposed system enhances security, minimizes fraud, and improves efficiency in document processing. Furthermore, it contributes to the advancement of artificial intelligence applications in cybersecurity and data validation.

CHAPTER 1: INTRODUCTION

I. Introduction

In today's digital era, documents such as certificates, identification cards, transcripts, and contracts are widely used for verification and authentication purposes. However, the rise of digital tools has also made it easier to manipulate or forge documents. This creates serious concerns in sectors like education, employment, and government services, where document authenticity is critical.

Traditional document verification methods rely heavily on manual inspection, which can be inefficient and unreliable, especially when dealing with large volumes of documents. Human evaluators may overlook subtle signs of forgery, leading to potential fraud.

Artificial Intelligence (AI) offers a promising solution to this problem. Through machine learning and pattern recognition, AI can detect inconsistencies and anomalies in documents that are difficult for humans to identify. This study aims to develop an AI-based document verification system capable of distinguishing between authentic and fake documents.

II. Significance of the Study

This system will benefit the following:

- **For Institutions:** Improves security and reduces fraud in document processing.
- **For Administrators:** Minimizes manual workload and increases efficiency.

- **For Researchers:** Provides a foundation for further studies in AI-based security systems.
- **For Students/Developers:** Enhances knowledge in AI, machine learning, and cybersecurity.

III. Statement of the Problem

1. How can an AI-based system be developed to detect fake or tampered documents?
2. What features (e.g., text patterns, signatures, layout) are most effective in identifying document authenticity?
3. How accurate and reliable is the proposed system compared to manual verification?
4. How can the system improve efficiency in document validation processes?

IV. Scope and Limitations

The system focuses on detecting fake documents using image and text analysis. It supports common document formats such as images (JPG, PNG) and PDFs. The system is limited by the quality and size of the training dataset and may not detect highly sophisticated forgeries. It does not replace legal verification processes but serves as a support tool.

OPERATIONAL DEFINITION OF TERMS

Artificial Intelligence (AI): A branch of computer science that enables machines to simulate human intelligence, particularly in learning and decision-making.

Machine Learning: A subset of AI that allows systems to learn from data and improve performance without being explicitly programmed.

Document Verification: The process of checking whether a document is authentic, valid, and not tampered with.

Fake Document: A document that has been altered, forged, or fabricated to misrepresent information.

Image Processing: A technique used to analyze and manipulate images to extract useful information.

Feature Extraction: The process of identifying important characteristics (e.g., text, patterns, layout) from a document for analysis.

Classification: A machine learning process of categorizing data into predefined classes (e.g., real or fake).

Dataset: A collection of data used to train and test the machine learning model.

Optical Character Recognition (OCR): Technology used to convert images of text into machine-readable text.

System Accuracy: The measure of how correctly the system identifies real and fake documents.